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What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 8: Convergence and Error Control

Group-B

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1 Why the finite starting state matters

Starting with uncoupled product columns at finite X is a controlled initial-state choice, not the exact mixed ultraviolet vacuum. The $r = 0.03$ control gives

Initial X	State	F after regulator extrapolation
800	product	-3.043×10^{-10}
800	adiabatic	-3.986×10^{-10}
1600	product	-4.811×10^{-10}
1600	adiabatic	-3.986×10^{-10}

The product-state change dominates the desired signal. To remove this leading ultraviolet mismatch, let $Y = (Q, \Sigma)^T$ and seek $Y' = R(x)Y$. The second-order equations imply

$$R' + R^2 - \frac{2I + gJ}{x}R + I - \frac{D}{x^2} = 0, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & -3g \\ 0 & -M^2 \end{pmatrix}. \tag{1}$$

Use $R = iI + \sum_{n=1}^{p} R_n/x^n$. Equating powers gives

$$R_1 = I + gJ/2, \tag{2}$$

$$2iR_n = [(n+1)I + gJ]R_{n-1} - \sum_{j=1}^{n-1} R_j R_{n-j} + \delta_{n2}D. \tag{3}$$

At initial X, the canonical relation is $\Pi = AY$ with $A = -R/X^2 + gE_{12}/X^3$. It is symmetric. Choose the position columns F with

$$FF^T = \frac{X^2}{2}(\operatorname{Im} R)^{-1}, \qquad \dot{Y} = -XRF. \tag{4}$$

Then $2\operatorname{Im}\langle Y\Pi^T\rangle = I$ initially. This defines a pure positive-frequency initial state. Orders 4/6 and $X = 100$–800 are compared in Step 4. It does not rescale a final covariance or impose the desired answer.

2 Regulator removal and the numerical error budget

For the production pilot we use order 6, $X = 1600$, $x_f = 10^{-6}$ and DOP853 tolerances $10^{-12}/10^{-15}$. At each r, all legs use the same physical final time. We evaluate $\epsilon = 0.1, 0.05, 0.025, 0.0125$ and compare linear and quadratic zero-regulator extrapolations of the last three points. At fixed regulator the prescribed $(\epsilon, z_{\max}) = (0.1, 80), (0.05, 160), (0.025, 320)$ pairs are tested again at doubled cutoff. The integration grid resolves the early oscillations with intervals no longer than $\pi/[4(2 + r)]$, with logarithmic integration below $z = 1$. Halving the intervals and increasing Gauss order from 12 to 16 tests quadrature. Independent adaptive Gauss–Kronrod integration checks $r = 0.03$, $\epsilon = 0.025$; its difference from the Gauss calculation is 1.676×10^{-22}. We separately vary $X = 800/1600$, ODE tolerances, quadrature, and $z_f = 0.01, 0.003, 0.001$. The conservative error is the sum of these changes and the fixed-regulator tail change. These are systematic numerical variations, not independent Gaussian standard deviations.

r	F extrapolated	Conservative error
0.2	1.812×10^{-8}	1.059×10^{-10}
0.1	1.585×10^{-9}	2.857×10^{-11}
0.03	-3.986×10^{-10}	1.104×10^{-12}
0.01	6.258×10^{-11}	2.940×10^{-13}
0.003	2.482×10^{-11}	4.818×10^{-14}
0.001	-2.966×10^{-13}	1.414×10^{-14}

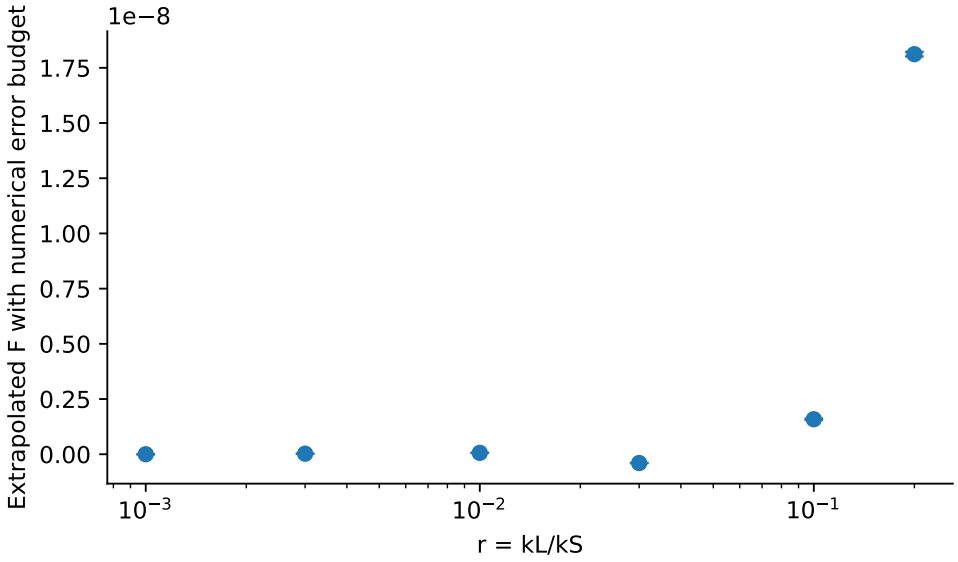


Figure 1: Six pilot triangles with the sum of tested numerical changes. A large relative error near a zero of the oscillation is distinguished from the uncertainty in its envelope.

The largest pilot error normalised to the clock-envelope scale is 0.005845. This supports a resolved shape scan. Step 9 checks the stronger condition against the fitted oscillatory component itself and tests background and fit-window dependence. **Code and supporting data:** step8 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Ultraviolet initial-state derivation and interpretation.
Samarth Bhatnagar	Regulator, cutoff and final-time error budget.
Mayank Sinha	Quadrature/precision comparisons and convergence data.